JIARUI (VERA) SHI

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EDUCATION

University of Pennsylvania, Philadelphia, PA — Aug 2025 – May 2027
Master of Science in Computer and Information Science
Selected Coursework: Algorithms, Databases, Trustworthy Machine Learning, Computer & Network Security

New York University Shanghai, Shanghai, China — Sept 2021 – May 2025
Bachelor of Science in Computer Science and Data Science (Concentration: Artificial Intelligence), Summa Cum Laude
Cumulative GPA: 3.93 / 4.00 Major GPA: 4.00 / 4.00

Study Abroad: New York University, New York, NY — Aug 2023 – May 2024

WORK EXPERIENCE

Amazon — Jun 2026 – Aug 2026
Software Development Engineer Intern — *Seattle, WA*

- Designed and implemented a multi-tenant, configuration-driven GenAI platform using React/TypeScript, Amazon Bedrock, AWS Lambda, DynamoDB, and SQS, enabling partner teams to self-serve new manager-facing AI insights through declarative configuration and reducing launch time from approximately 3 weeks to under 1 day (>95%).
- Built an event-driven distributed serverless pipeline spanning data ingestion, orchestration, model inference, and insight publishing, achieving sub-5-second p99 generation latency through concurrency control and DLQs; provisioned infrastructure with AWS CDK and executed zero-downtime DynamoDB GSI migration.
- Developed an automated GenAI quality-assurance pipeline combining an LLM-as-a-judge evaluation system, deterministic grounding checks, prompt-injection isolation, and PII/output validation, suppressing over 80% of low-quality or hallucinated outputs before production publication.

Knova — Feb 2025 – Jul 2025
Full Stack Developer Intern (Early-stage AI startup headquartered in Palo Alto, CA) — *Shanghai, China*

- Built an AI agent operations console for Tapi, Knova's AI product, enabling content and operations teams to configure, deploy, and monitor production agents through versioned prompts, data-source configurations, and publishing policies.
- Developed a LangGraph-based agent execution pipeline with asynchronous job handling, persistent run state, automated output validation, and bounded revision/replanning loops, routing failed outputs through targeted recovery paths before escalating to human review.
- Implemented versioned agent and prompt configurations, evaluation traces, and end-to-end audit logging using Next.js, React, TypeScript, REST APIs, MongoDB, S3, and AWS EC2; reduced routine content turnaround from ˜3h to <20 minutes while supporting 80+ production agents and 320+ daily runs.

CoStone Capital — Jun 2024 – Aug 2024
Full Stack Developer Intern — *Shanghai, China*

- Extended Dash-based Gflows for dynamic ticker management, synchronized across clients with enhanced update frequency, and automatically deployed on a server using GitHub Actions, integrating Polygon.io's data feed via RestClient.
- Developed a market data visualization and trading tool using lightweight-charts-python and PyQt5, enabling real-time data visualization and trade execution via IB Trader Workstation API and Polygon.io as the data source.

SELECTED RESEARCH EXPERIENCE

LLM Alignment for Sycophancy Mitigation — Feb 2026 – May 2026
Trustworthy Machine Learning — *University of Pennsylvania*

- Implemented a modular LLM alignment pipeline for automated dataset construction, behavioral evaluation, and fine-tuning, integrating GPT-4.1 Nano with SFT/DPO training and GPT-4.1 Mini for scalable preference-data synthesis.
- Built a scalable data generation and validation workflow producing 2.8K+ high-quality adversarial training conversations for alignment through answer verification and scalable preference-pair synthesis.
- Developed a behavioral evaluation framework measuring Sycophancy Rate (SR), Accuracy Drop (AD), and Legitimate Correction Rate (LCR), and conducted controlled dataset ablations and hyperparameter tuning to reduce sycophantic behavior by 77% (SR: 15.2% → 3.5%) while improving LCR from 90.4% to 94.9%.

SKILLS

Languages: Python, Java, Go, C/C++, JavaScript/TypeScript, SQL, x86-64 assembly (Intel syntax), Verilog, Swift
Full Stack: Node.js/Express, FastAPI, REST APIs, JWT/OAuth2, Django, React, Next.js, Tailwind CSS, HTML/CSS
ML/AI: PyTorch, scikit-learn, NumPy/pandas, torchaudio/librosa, Weights & Biases, Hugging Face Transformers
Agentic AI: LangChain, MCP, tool calling, Retrieval/RAG, LLM evaluation, Amazon Bedrock/AgentCore
Data/Cloud: MongoDB, PostgreSQL, Redis; AWS (Bedrock, Lambda, EC2, S3), Docker, NGINX, GitHub Actions